

V4.2 Toy Model 5F

Dark Sector Track Close-Out and Observational Guardrail Map

Rev A - Consolidation of Toy Models 5A-5E

Date: 2026.05.26 | Continuation after Toy Model 5E

Status: Toy Model 5F closes the exploratory Dark Sector track by separating derived results, assumed rules, and untested observational requirements. It does not claim that dark matter, dark energy, Lambda-CDM, MOND, or physical 3T have been solved or replaced.

V4.2 Dark Sector Track 5A-5E Progression

5A	Rotation-curve analogue	Flat curve produced
↓		
5B	Graph edge economics	$a_{rel} \sim 1/r$ and $M_{rel} \sim r$ derived
↓		
5C	BTF exponent check	constant Gamma fails; square-root law required
↓		
5D	Universal threshold A_R	$T_{edge}^2 = A_R G M_b$ derived from capacity action
↓		
5E	Local-global bridge	$p_C = -\rho_C$; $w = -1$ analogue

1. Executive Close-Out

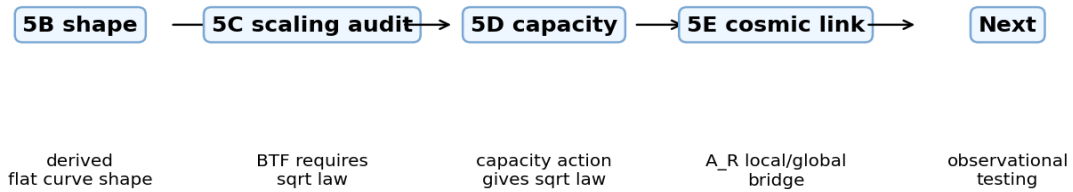
Toy Models 5A-5E built a dark-sector analogue sequence on top of the closed weak-field gravity baseline from 4A-4J. The sequence shows how relational network economics can mimic two major dark-sector signatures: flat galactic rotation curves and a dark-energy-like background pressure.

The strongest defensible conclusion is limited: the framework achieved structural compatibility with selected dark-sector signatures in toy models. It did not achieve physical validation against real observations. The track is therefore closed as an exploratory mathematical baseline, not as a completed cosmological theory.

2. Model-by-Model Audit

Model	Target	Input	Result	Close-out status
5A	Rotation-curve analogue	Assumed relational-tension term	Flat curve produced	Pass as analogue; term was inserted
5B	Graph edge economics	2D annular radial action	$a_{rel} \sim 1/r$ and $M_{rel} \sim r$ derived	Shape derived, amplitude still open
5C	BTF exponent check	Mass-scaling audit	constant Gamma fails; square-root law required	Critical diagnostic success
5D	Universal threshold A_R	Capacity action	$T_{edge}^2 = A_R G M_b$ derived from capacity action	A_R/x_i not physically derived
5E	Local-global bridge	A_R as local BTF scale and global capacity floor	$p_C = -\rho_C$; $w = -1$ analogue	Cosmology data not tested

Dark Sector Development Sequence



3. Derived, Assumed, and Open Pieces

3.1 Derived or structurally established pieces

Piece	Status	Reason	Guardrail
Flat outer rotation shape	Derived in 5B	From 2D annular radial graph action and edge-flux conservation.	Conditional on effective graph dimension $D_{\text{eff}} = 2$.
1/r relational acceleration	Derived in 5B	Outside source, $r \, d\Phi/dr = \text{constant}$, so $ d\Phi/dr \sim 1/r$.	Requires acceleration to follow edge-response gradient.
M_{rel} proportional to r	Derived in 5B	If v_{rel}^2 is constant, Newtonian-inferred $M_{\text{rel}} = v^2 \, r/G$.	Equivalent-mass interpretation, not literal mass.
Constant coupling fails BTF	Derived in 5C	$v_{\text{flat}}^2 = \Gamma M_b \rightarrow v_{\text{flat}}^4 \sim M_b^2$.	Important negative result.
$\Gamma(M_b) \sim M_b^{-1/2}$ required	Derived in 5C	General $\Gamma \sim M^q$ gives exponent $2(1+q)$; BTF requires $q=-1/2$.	Identifies required scaling.
Capacity action gives square-root law	Derived in 5D	$L = T^2/(2A_R) - GM \ln(T/T_{\text{ref}}) \rightarrow T^2 = A_R G M$.	Depends on chosen capacity action.
Local-global A_R formal bridge	Structured in 5E	$A_R = \xi \, c \, H_C$ can be used locally and globally.	ξ not derived; physical value not fixed.
$w=-1$ capacity background	Derived in 5E	ρ_C constant $\rightarrow p=-\rho_C$.	Toy cosmological-constant analogue only.

3.2 Assumed or imported pieces

Assumption	Where	Why it was used	Required future work
Effective 2D annular graph at galactic scale	5B	Explains flat shape if true.	Needs physical/geometric justification.
Relational acceleration follows edge-response gradient	5B	Maps graph response to circular velocity.	Needs deeper metric/dynamics derivation.
Universal capacity action form	5D	Produces square-root BTF law.	Must be derived from deeper 3T/graph principle.
$A_R = \xi \, c \, H_C$	5D/5E	Links local galaxy scale to cosmic horizon scale.	ξ and physical normalization not derived.
Constant capacity density	5E	Produces $p=-\rho$, $w=-1$ analogue.	Needs real expansion-history testing.
Dimensionless toy normalization	5A-5E	Keeps structure visible.	Physical units and constants not established.

The most important open assumptions are the effective 2D annular nature of galaxy-scale relational tension, the capacity-action form, and the physical normalization of A_R and ξ .

4. Mathematical Backbone of the Closed Track

4.1 Flat-curve shape from 5B

$$A[\Phi] = 2 \pi \int r \left[\frac{1}{2} \left(\frac{d\Phi}{dr} \right)^2 - S(r) \Phi(r) \right] dr$$

Variation:

$$\left(\frac{1}{r} \right) \frac{d}{dr} \left[r \frac{d\Phi}{dr} \right] = -S(r)$$

Outside source:

$S \approx 0$

$$r \frac{d\Phi}{dr} = \text{constant}$$

$$|d\Phi/dr| \propto 1/r$$

If a_{rel} follows $|d\Phi/dr|$:

$$a_{\text{rel}} \propto 1/r$$

$$v_{\text{rel}}^2 = r a_{\text{rel}} = \text{constant}$$

$$M_{\text{rel}} = v_{\text{rel}}^2 r / G \propto r$$

4.2 BTF requirement from 5C

If:

$$v_{\text{flat}}^2 = \Gamma M_b$$

constant Γ gives:

$$v_{\text{flat}}^4 = \Gamma^2 M_b^2$$

This fails BTF.

General:

$$\Gamma(M_b) \propto M_b^q$$

$$v_{\text{flat}}^4 \propto M_b^{2(1+q)}$$

BTF requires:

$$q = -1/2$$

4.3 Capacity action from 5D

$$L(T; M_b) = T^2 / (2 A_R) - G M_b \ln(T/T_{\text{ref}})$$

$$dL/dT = T/A_R - G M_b/T = 0$$

therefore:

$$T^2 = A_R G M_b$$

Since $T = v_{\text{flat}}^2$:

$$v_{\text{flat}}^4 = A_R G M_b$$

4.4 Local-global link from 5E

Local:

$$v_{\text{flat}}^4 = A_R G M_b$$

Global:

$$A_R = \xi c H_C = \xi c^2 / R_H$$

Capacity background:

$$\rho_C = \text{constant}$$

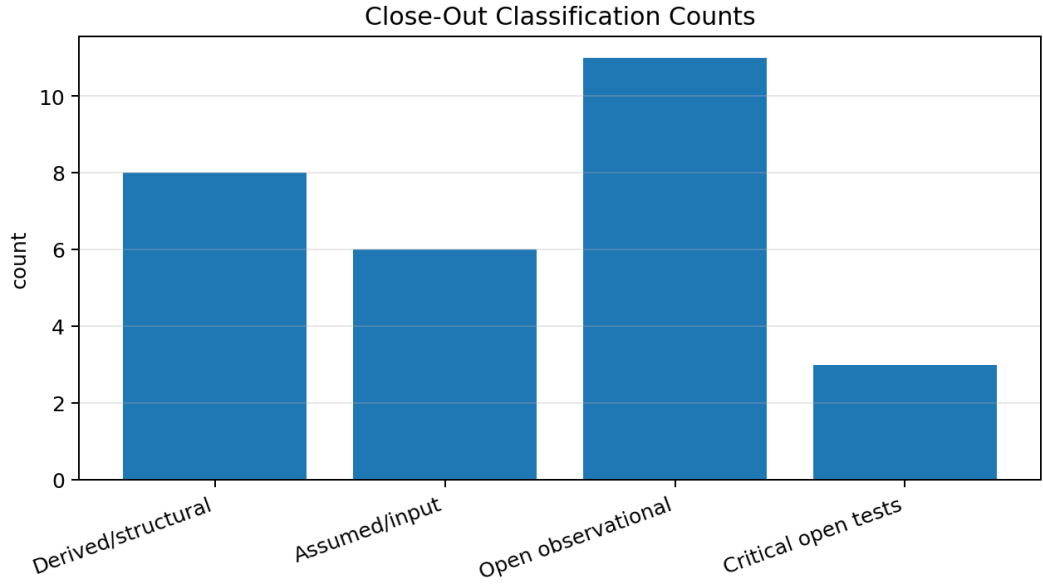
$$p_C = -\rho_C$$

$$w_C = -1$$

5. Numerical Checkpoints

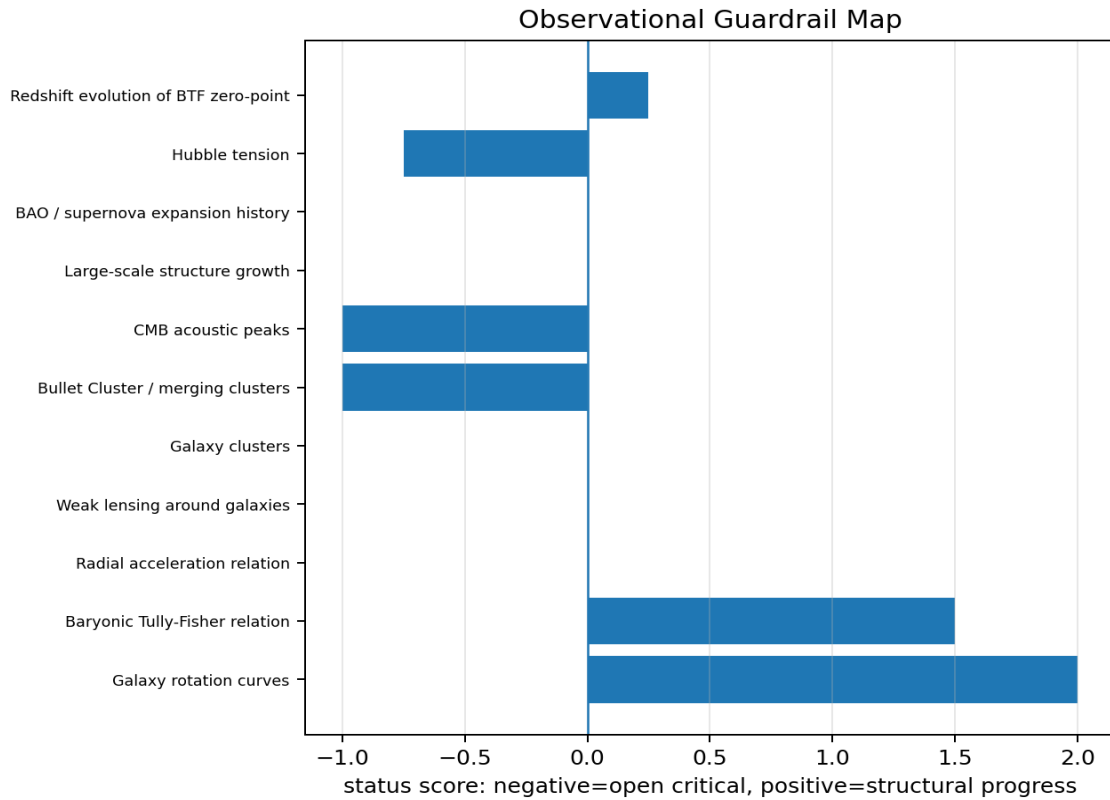
Model	Diagnostic	Value	Meaning
5A	outer visible-only velocity slope	-0.4909	Kepler-like decline
5A	outer total velocity slope	-0.0312	flat with assumed relational tension
5A	equivalent density slope	-2.1043	halo-like profile
5B	Euler residual max	8.886e-06	radial graph equation solved
5B	outer edge-flux CV	3.969e-07	outer flux effectively constant
5B	outer relational velocity slope	0.0000	flat contribution derived
5B	outer M_{rel} slope	1.0000	linear inferred mass
5C	linear coupling BTF slope	2.000000	constant coupling fails BTF
5C	threshold BTF slope	1.000000	square-root law passes exponent
5C	required Γ slope	-0.500000	Γ must scale as $M_b^{-1/2}$
5D	capacity/BTF slope	1.000000	capacity law gives BTF exponent
5D	dL/dT residual max	1.421e-14	capacity equilibrium solved
5D	v vs A_R slope	0.250000	v_{flat} proportional to $A_R^{1/4}$
5E	BTF slope retained	1.000000	local threshold retained

Model	Diagnostic	Value	Meaning
5E	xi required	0.361557	normalization factor not derived
5E	q_today	-0.549986	accelerating toy background
5E	p+rho max	0.000e+00	checks w=-1 analogue



6. Observational Guardrail Map

This is the most important output of 5F. It marks where the V4.2 Dark Sector track currently stands against the observational targets that any serious dark-sector framework must address.



Observational target	Current V4.2 status	Required test	Gap before physical claim
Galaxy rotation curves	Structurally compatible via 5A-5B	Fit real curves across radii with baryons only + relational term.	Requires galaxy dataset, baryonic mass models, uncertainty handling.
Baryonic Tully-Fisher relation	Conditionally compatible via 5C-5D	Recover slope, zero-point, and scatter without tuning per galaxy.	A_R physical value and scatter mechanism open.
Radial acceleration relation	Open	Reproduce observed acceleration relation from baryonic distribution.	Requires acceleration-level comparison, not just velocity curves.
Weak lensing around galaxies	Open	Predict lensing mass profile matching rotation-derived profile.	Requires relativistic lensing extension of 5B/5D.
Galaxy clusters	Open	Explain cluster dynamics and lensing without arbitrary hidden mass.	Requires 3D/cluster-scale graph geometry.
Bullet Cluster / merging clusters	Open critical test	Separate lensing peaks from baryonic plasma correctly.	Requires dynamical collision model and lensing map.
CMB acoustic peaks	Open critical test	Reproduce early-universe power spectrum effects attributed to dark matter/dark energy.	Requires relativistic cosmology and perturbation theory.
Large-scale structure growth	Open	Reproduce growth rate and matter power spectrum.	Requires cosmic perturbation dynamics.
BAO / supernova expansion history	Open	Fit distance-redshift observables with capacity background.	Requires calibrated $H(a)$, not toy $H(a)$.
Hubble tension	Not addressed	Explain or not worsen current H_0 discrepancies.	Requires real late/early-universe analysis.
Redshift evolution of BTF zero-point	Future discriminant	Check whether A_R is tied to residual floor H_C or time-varying $H(a)$.	Potential observational falsifier.

7. Strongest Defensible Claim

The strongest defensible claim after 5A-5F is:

Within controlled toy models, the V4.2 relational framework can reproduce the structural form of flat galactic rotation curves, identify the BTF scaling requirement, express that scaling through a universal capacity action, and connect the same capacity scale to a dark-energy-like constant background pressure. This is structural compatibility, not observational validation.

8. Claims Explicitly Not Made

- No claim that particle dark matter is false.
- No claim that Lambda-CDM is replaced.
- No claim that MOND is derived or replaced.
- No claim that real galaxy data has been fitted.
- No claim that weak lensing, clusters, Bullet Cluster, CMB, BAO, supernovae, or structure formation have been explained.
- No claim that the physical value of A_R, xi, H0, Lambda, or any acceleration scale has been derived.
- No claim that physical 3T has been proven.
- No claim that the Dark Sector is solved.

9. Critical Risks and Falsification Routes

Risk	Consequence	Required response
Effective 2D graph assumption fails	Flat rotation shape may not be native.	Derive graph dimension from galaxy formation / 3T topology.
A_R not universal	BTF zero-point fails.	Test scatter and redshift evolution.
Lensing mismatch	Rotation analogue insufficient.	Build relativistic lensing extension.
Clusters fail	Theory cannot replace dark matter broadly.	Cluster dynamics and lensing map tests.
CMB peaks fail	Cosmological theory invalid as dark-sector replacement.	Full perturbation theory required.
Structure formation fails	Global capacity background insufficient.	Growth-rate and power-spectrum tests.

10. Recommended Next Work Sequence

The correct next step is not to declare victory. The correct next step is to move from toy-model compatibility to the first real falsifiable data interface.

- 5G - Minimal Real-Galaxy Rotation-Curve Fitting Protocol: choose one public galaxy dataset and define the fitting variables without hiding extra freedom.
- 5H - BTF Zero-Point and Scatter Test: test whether one universal A_R can fit a family of galaxies.
- 5I - Relativistic Lensing Extension: derive the lensing prediction from the same metric response.
- 5J - Cluster Collision Guardrail: formulate what the model would need to reproduce a Bullet-Cluster-like separation.
- 5K - Cosmological Perturbation Roadmap: identify the equations needed before touching CMB or structure formation.

Recommended immediate continuation: create the real-data protocol first, before fitting anything. That prevents accidental curve-fitting.

11. Rebind Prompt for Continuation

We are resuming after Toy Model 5F.

5F closed the exploratory Dark Sector track 5A-5E.

5A:
Assumed relational-tension term and showed flat rotation-curve analogue.

5B:
Derived flat rotation shape from effective 2D annular graph action:
 $A[\Phi] = 2 \pi \int r [1/2 (d\Phi/dr)^2 - S \Phi] dr$
 $\Rightarrow (1/r) d/dr [r d\Phi/dr] = -S$
Outside source: $r d\Phi/dr = \text{constant}$
 $\Rightarrow a_{rel} \sim 1/r$
 $\Rightarrow v_{rel}^2 \sim \text{constant}$
 $\Rightarrow M_{rel} \sim r$

5C:
Constant coupling fails BTF:
 $v_{\text{flat}}^2 = \Gamma M_b \rightarrow v_{\text{flat}}^4 \sim M_b^2$
BTF requires $\Gamma \sim M_b^{-1/2}$
or $T_{\text{edge}}^2 = A_R \Gamma M_b$.

5D:
Capacity action:
 $L(T; M_b) = T^2 / (2A_R) - G M_b \ln(T/T_{\text{ref}})$
Variation gives:
 $T^2 = A_R \Gamma M_b$
Since $T = v_{\text{flat}}^2$:
 $v_{\text{flat}}^4 = A_R \Gamma M_b$

5E:
 A_R linked structurally to residual cosmic capacity:
 $A_R = \xi c H_C = \xi c^2 / R_H$
 ρ_C constant $\rightarrow p_C = -\rho_C \rightarrow w = -1$

5F close-out:
Derived/structural:

- flat curve shape in effective 2D graph
- BTF required scaling identified
- capacity action produces square-root law
- local/global A_R bridge structurally written

Assumed/open:

- effective 2D galactic relational graph
- capacity action form
- ξ and physical A_R value
- real data, lensing, clusters, CMB, structure formation

Strongest defensible claim:
V4.2 is structurally compatible with selected local/global dark-sector signatures in toy models.

Do not claim:
dark matter solved, dark energy solved, Lambda-CDM replaced, particle dark matter disproved, physical 3T proven.

Recommended next:
5G - Minimal Real-Galaxy Rotation-Curve Fitting Protocol,
or prepare a wider V4.2 binder before moving to real data.

12. Final Close-Out Statement

Toy Model 5F seals the Dark Sector toy-model track. The work achieved a coherent structural chain from galaxy-scale graph economics to a local BTF-like threshold and then to a global dark-energy-like capacity background. The track is ready to be preserved as a binder baseline. It is not ready to be presented as a physical replacement for dark matter or dark energy until the observational guardrail map is confronted.